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ThoxWork: One Workplace, One Connected Team. A Policy-Bounded DigitalHuman Work Platform for Enterprise Use

ThoxWork is the third THOX platform: the work-organization layer in which a company assigns real work to named DigitalHuman colleagues (ThoxEmployees), reaches them through Rolodex, bounds their authority and resources by MeshGroup, and draws on approved capacity through MeshStack. This paper defines the model formally, reports what is actually built, and evaluates the model's five load-bearing invariants in a deterministic virtual lab.

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ABSTRACT

Enterprise adoption of AI agents is outrunning enterprise governance of them. Vendor surveys report organizations experimenting with agents at scale while only a minority operate agent policies, and recent orchestration research shows that at team scale the deciding factors are not model capability but discovery scope, handoff fidelity, and the structure of human review. THOX's answer is a work platform, not another agent framework. In the ThoxWork model, work lives in shared workspaces; DigitalHumans are named, role-holding colleagues who can be assigned tasks and held to account for them; Rolodex is the team interface through which a person assigns a task, checks in, and reviews results; a MeshGroup binds a named team to the exact set of resources it is approved to use; and MeshStack connects those approved resources. This paper makes four contributions. First, it states the model as five testable invariants: unapproved resources never run, oversized work is denied rather than spilled, reassignment preserves a complete handoff chain, no task reaches Done without eligible human review, and queues complete predictably as teams and groups scale. Second, it reports an evidence-bounded account of what THOX has actually built toward the model, separating shipped and measured components from designed and proposed ones. Third, it evaluates all five invariants in a deterministic, seeded virtual lab (20 replicates per experiment; 320,000 simulated tasks in the scale grid alone), in which every pre-registered bar is met and every number is labelled SIMULATED. Fourth, it states the model's limits in the same terms: the lab proves the rules as modelled, not a production deployment, and the falsifiable content of the scale result is non-collapse, not speedup. ThoxWork as a purchasable enterprise platform remains in development; THOX's own product copy says so, and this paper claims nothing further.

0 / 12,000
Unapproved-resource attempts leaked across direct, cross-group and escalation attacks (E1)
SIMULATED

1.000
Handoff-chain completeness over 5,000 reassigned tasks and 15,000 chain entries; 0 of 4,000 negative probes leaked (E2)
SIMULATED

100%
Of 4,000 oversized tasks denied UNPLACEABLE; 0 silently placed, 0 spilled to another group (E3)
SIMULATED

0 / 6,000
Review-gate bypasses succeeded across six attack classes; 4,000 of 4,000 legitimate tasks completed through review (E4)
SIMULATED

Status at writing, 2026-11-04. This is edition v1.0.1, the public edition under the THOX Research Paper Process v1.0. The system-model and product-status claims rest on THOX product records identified in the provenance register (Section 12). The evaluation claims rest on a single evidence source, `exp/results/summary.json`, produced by a deterministic simulator and re-verified by a reviewer who did not produce it: two independent runs of the suite produce a byte-identical summary (SHA-256 `f7c72f64b9b8c…`). Nothing in this paper is measured on a device, a network, or a customer deployment, and no customer, deployment count, or commercial term is claimed anywhere in it.

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1 Introduction: the third platform

THOX builds local-first AI in layers. Its devices and operating system put capable models on hardware the customer owns. Its MeshStack fabric lets approved devices pool their capacity for each other, under grant-gated, lease-fenced rules that have been published and measured elsewhere [15]. Neither layer, by itself, answers the question a company actually asks on a Monday morning: who is doing this work, what are they allowed to touch, and how do I know it is done?

ThoxWork is the layer that answers that question. It is THOX's third platform: the work-organization and governance layer for companies that employ DigitalHumans alongside people. Its positioning, fixed in THOX product records in October 2026 and reaffirmed in November 2026 planning, is one sentence long: One workplace. One connected team. ThoxWork organizes the work. Rolodex connects you to your DigitalHuman team. MeshGroups define team access. MeshStack connects approved resources.

The vocabulary matters, because each term names a distinct responsibility that other approaches merge or omit:

- **DigitalHumans** are named, role-holding synthetic colleagues. A DigitalHuman has an identity, a role, a team, and a bounded remit. Under THOX's DigitalHuman Colleague and Disclosure Standard, they are treated and credited as colleagues, not tools.
- **ThoxEmployees** is the workforce offering built on DigitalHumans: digital employees an organization staffs into roles, as distinct from a chatbot a person prompts.
- **Rolodex** is the team interface: the place a person finds their DigitalHuman team, assigns a task, checks in on it, and reviews the result.
- **A MeshGroup** is the unit of bounded authority: a named set of team members plus the exact set of resources that team is approved to use.

- **MeshStack** is the resource fabric a MeshGroup draws on: approved models, files, projects and results, connected under the published MeshStack rules.
- **ThoxWork** is the platform that holds these together: shared workspaces, assignments, policy, and the evidence trail of who did what, reviewed by whom.

This paper is deliberately narrow about evidence. Section 4 separates what is built from what is designed and what is proposed, product by product, because a platform paper that blurs those lines teaches its readers to discount everything else in it. Sections 5 and 6 evaluate the rules of the model in a virtual lab that anyone can re-run: the lab is a deterministic simulation, its outputs are labelled SIMULATED throughout, and its pre-registered pass bars were fixed before the first run. Where the lab's own construction produces an artifact that could be misread as a result, Section 7 says so explicitly.

2 The enterprise problem

Four findings from the recent literature frame the design problem ThoxWork addresses. None of them evaluates ThoxWork; each is used here as a design warrant, and the distinction is maintained throughout.

Agents are scaling faster than their governance. McKinsey's State of AI 2025 (N = 1,993, 105 countries) reports 62% of organizations at least experimenting with agents, but only 23% scaling agents in at least one function and at most 10% in any given function [10]. A SailPoint survey of IT professionals (N = 353, May 2025) reports 82% of organizations running agents while only 44% have policies governing them. The gap between deployment and governance is the normal state of the market, not an exception.

At team scale, orchestration structure dominates model capability. Dhanyamraju, Raghav and Lee study autonomous multi-agent orchestration across 208 production-derived scenarios at scales of under 10, 20 to 80, and 200 agents, and find that scale, not task complexity, dominates failure, largely through agent-discovery noise; a task manager with priority, merging and preemption cuts high-priority latency by 14 to 75% and raises related-event correctness by 20 points [1]. The design implication is that discovery and assignment must be scoped to a team, not broadcast across an unbounded fleet.

Handoffs are where multi-agent work loses information. Liu, Li, Wang and Wang train only the orchestrator in DynaFlow to emit executable handoff programs, and reach 88.08% across six benchmarks, 10.9 points above the best baseline on the same base model; cross-stage information flow is the deciding factor [2]. Free-form agent chat is a weak handoff medium. An explicit, inspectable handoff artifact is a stronger one. (The benchmark figure is a reasoning-benchmark result; this paper does not transfer it to enterprise tasks.)

The human-agent workspace itself is unspecified by current protocols. The Model Context Protocol standardizes agent-to-tool access and the A2A protocol standardizes agent-to-agent delegation [13]. Shahid, Suttie and Black argue in the CHAP proposal that neither specifies the accountable human-

agent workspace: shared tasks and artifacts, structured overrides, portable handoffs, and signed approvals over an append-only evidence log [3]. CHAP is the closest published analogue to the ThoxWork team layer, and Section 8 compares the two directly.

Two further anchors shape the workforce framing. Weill and Woerner at MIT CISR define the digital colleague as an agent that is autonomous, in dialogue, continuous, auditable, and that escalates consequential decisions, and is therefore managed like an employee; in their survey (N = 132), value concentrates in the under-20% of firms that redesign workflows, roles and governance around such colleagues [4]. Microsoft's 2026 Work Trend Index (N = 20,000) reaches a parallel conclusion: organizational factors account for roughly twice the outcome variance of individual factors, and the professionals who get the most from agents are those who build multi-agent workflows and set team standards [9]. Finally, Ju and Aral's field experiment (2,310 participants, 11,138 advertisements, roughly 4.9 million impressions) provides the strongest causal evidence that human-AI teams working inside a shared workspace produce substantially more output per worker, alongside a measurable change in the social texture of the work itself [5]. A platform for agent work must therefore ship an operating model, not only agents. That is the problem statement for Section 3.

3 The ThoxWork model

The model is stated here at the level of rules, not implementation. Implementation mechanisms that are not already published are THOX trade secrets and are out of scope for this paper; the rules below are the public contract the platform is evaluated against in Sections 5 and 6.

FIGURE 1 · THE THOXWORK STACK. EACH LAYER HAS ONE RESPONSIBILITY; AUTHORITY NARROWS DOWNWARD, EVIDENCE FLOWS UPWARD.

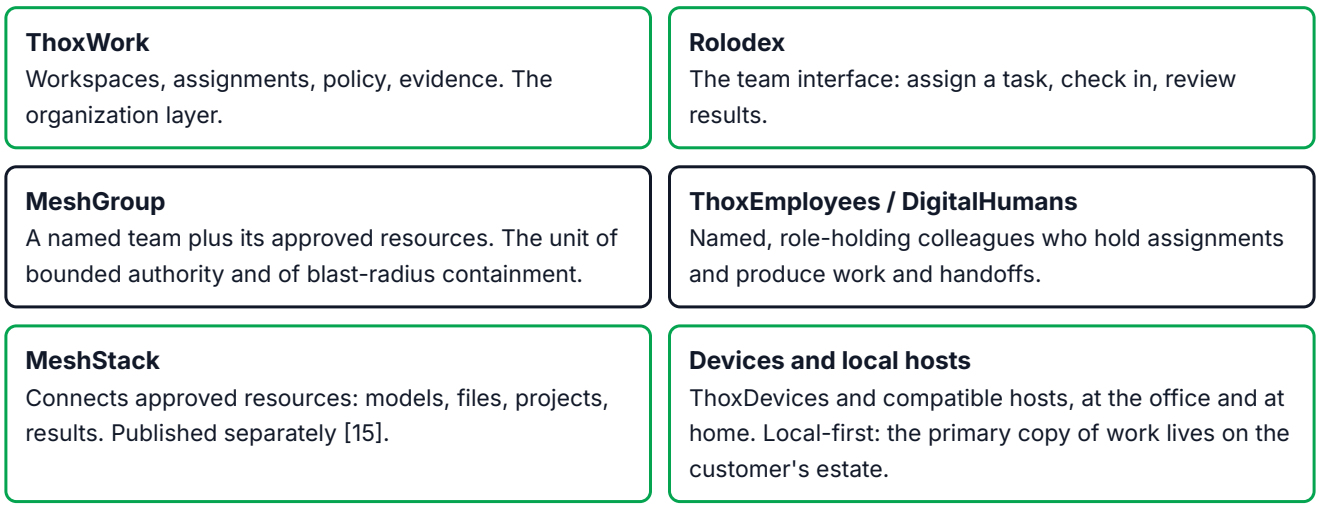


Figure 1. Layer responsibilities in the ThoxWork model. The diagram states design positions, not deployment measurements.

3.1 Definitions

Definition 1 (MeshGroup). A MeshGroup G is a pair (M, R) of a member set M (people and DigitalHumans) and an approved resource set R. The pooled approved capacity of G is the sum of the capacities of the resources in R. There is no global pool: capacity, membership and approval are properties of a group, never of the platform as a whole.

Definition 2 (Task). A task belongs to exactly one MeshGroup, declares the resources it requires and an abstract size (its demand), and carries exactly one assignee at a time. A task's required resources and demand are part of its record and are visible to its group.

Definition 3 (Handoff chain). Every reassignment of a task appends exactly one entry to its handoff chain: sequence number, previous holder, new holder, time, a non-empty written handoff note, and a hash of that note. Entries are append-only. A chain is complete when sequence numbers are contiguous, each entry's new holder is the next entry's previous holder, the final holder is the current assignee, and every note is present and hashes correctly.

Definition 4 (Legitimate completion). A task is legitimately Done when its record contains a review event in which an eligible reviewer, who is a member of the task's MeshGroup, is not the assignee, and holds a reviewer or manager role, approved the task while it was In Review, and the task entered Done in that same transition.

Definition 5 (Admission). A task is admitted to run only under the admission rule of Algorithm 1, evaluated when the task is created and again when it is assigned, so that approval cannot be checked once and exploited later.

Algorithm 1 · MeshGroup admission (public rule)

```
admit(task, group, free):
  1. Policy:       if task.required is not a subset of group.approved:
                       return DENIED, reason UNAPPROVED_RESOURCE
  2. Placeability: if task.demand > pooled_capacity(group):
                       return DENIED, reason UNPLACEABLE
  3. Current fit:  if task.demand > free:
                       return WAIT        (stays Queued; never placed elsewhere)
  4.               return ADMITTED

There is no partial placement, no demand splitting, no borrowing from
another group, and no fallback group. A DENIED task is terminal and is
shown, with its reason, in the group's Denied tray.
```

3.2 The Rolodex workflow

Rolodex presents work as four user-facing states with one visible exception state. The user's loop is three verbs: assign a task, check in, review results. Check-in is a progress act inside In Progress; it never advances a task to Done and never substitutes for review. Reassignment is permitted from Queued and In Progress, never from In Review, Done or Denied, and always requires the written handoff note of Definition 3.



Figure 2. The Rolodex state machine. Request-changes returns a task from In Review to In Progress. Denied is reachable only from admission (Algorithm 1) and is terminal.

3.3 The five invariants

The model stands or falls on five claims, each stated so it can be falsified:

1. **I1, policy enforcement.** A task that requires a resource its MeshGroup has not approved can never run with that resource attached, directly, through another group's approval, or by adding the resource after admission.
2. **I2, handoff integrity.** Every reassignment preserves a complete handoff chain (Definition 3); silent reassignment is impossible, including reassignment with an empty note, to a non-member, or out of In Review.
3. **I3, admission honesty.** Work larger than its group's pooled approved capacity is denied UNPLACEABLE in the tick it is evaluated. It is never placed partially, split, run on another group's resources, or left queued forever to disguise the denial.
4. **I4, the review gate.** No task reaches Done except by legitimate completion (Definition 4). Convenience paths, self-review, wrong-role review, non-member review and stale approvals all fail.
5. **I5, predictable scale.** Under a stated simulated-latency model, queues complete as team size and MeshGroup count grow; throughput does not collapse. I5 is a claim about the model's behavior under its stated assumptions, and Section 7 is explicit about how much weight it can bear.

4 What is built: an evidence-bounded account

This section reports the state of the THOX portfolio against the model, from a full audit of THOX repositories and product records conducted for this paper (44 GitHub repositories and 9 local product records; audit method in Section 12). Each component is labelled **Built** (code exists and is evidenced), **Designed** (specified, partially prototyped) or **Proposed** (positioning and plans only). Where THOX's own materials limit a claim, the limit is quoted rather than smoothed over.

Component	Status	Evidence of record	Limit carried with the claim
RolodexGo (Rolodex client, iOS/macOS)	Built	The shipped client is the 11-file contacts client on the repository main branch (conversation, DigitalHuman profiles, approval inbox, and Sadie Link pairing); the application carries an Xcode Organizer upload receipt (version 1.0.0, build 3). The Teams and Work tabs exist as a combined changeset of 2026-10-01: a Teams tab implementing ThoxEmployee team features (named teams from the DigitalHuman roster, the Queued → In Progress → In Review → Done queue, a written handoff note recorded on every reassignment) and a Work tab implementing the ThoxWork integration catalog (ThoxRoute, ThoxLLM, MeshStack, an MCP tool catalog, and the Sadie Link pairing), with 11 unit tests over the team models. The changeset was built and is test-covered, but it never landed in the repository main branch or in the build tree; it is preserved as an unlanded changeset, and whether the uploaded 1.0.0(3) build included it is unverified, so this paper does not claim that it did.	Local-first client. The claim for the shipped client is the contacts client plus its upload receipt, not live remote enterprise execution; the native preview is unpaired. In the changeset, team and work state are held on device and connection states in the Work tab are honest by construction, with integrations reporting unconfigured until configured. The tabs land in the shipped client as part of the ThoxWork implementation now underway.
ThoxEmployee (employee operating layer)	Built	An employee kernel with fail-closed policy, an operator action ceiling, digest-bound execution with preview, a persisted audit trail and hashed handoffs; pre-releases workspace-v0.3.0 and migration-studio-v0.2.0.	Pre-release software. The kernel evidences the bounded-authority half of the model; it is not, by itself, the ThoxWork platform.
MeshGroup	Designed, with a measured reference implementation of its policy core	MeshGroup is a canonical product concept in THOX records (team membership plus approved resources, connected through MeshStack). Its placement and policy core exists as a reference scheduler that passes 8 of 8 tests including cross-site default-deny and consent-gated placement scenarios, and as a model-configuration kit for enterprise workgroups.	There is no standalone MeshGroup repository and no distributed production backend claimed. The kit's models are published configurations on upstream weights and are explicitly not THOX-trained. Group isolation in those configurations is a prompting and routing convention, and the paper's I1 to I3 claims rest on the lab evaluation of the rules (Section 6), not on the kit.

Component	Status	Evidence of record	Limit carried with the claim
MeshStack	Built and published	Research paper v1.0.1, published with a reproducibility artifact [15]; backend test suites recorded at 241 of 241 and 25 of 25 smoke tests in THOX records.	Cited, not re-disclosed: this paper relies on the published MeshStack record and does not restate its internals. The corrected KV-compression range from the MeshStack v1.1.0 evidence work (1.17 to 1.20x on real GGUF workloads) supersedes the earlier 1.51x figure, which no THOX surface should use.
ThoxWork platform (hosted workspaces and enterprise execution)	Proposed, in active development	A monorepo in active development, a production web surface, a desktop workspace and the ThoxWork Creative Suite v0.1.0 (seven Rust applications, 2,449,152 bytes total in headless form, a zero-dependency core, 22 tests, and real demo outputs preserved as evidence) as a related workstream.	THOX's canonical product copy describes enterprise remote execution and the purchasable platform as Coming Soon, and the current web surface is a demonstration surface, not live enterprise execution. This paper's claims about ThoxWork as a platform are claims about the model and its lab evaluation, scoped exactly to that.
DigitalHuman roster and agent fleet	Built	A nine-role DigitalHuman roster operates in THOX's own work today under the Colleague and Disclosure Standard, and the ThoxAgent fleet (v1.2.0 line) evidences deterministic allocation and queued-not-dropped dispatch in an independent code audit (9 of 9 tests).	Internal operation evidences the roster and roles in THOX's own work. It is not a customer deployment claim, and none is made.

Table 1. Component status against the ThoxWork model. Status labels are the paper's own, assigned from the audit evidence, and are deliberately conservative.

Two audit findings deserve plain statement because they bound everything that follows. First, the strongest single integration evidence is the RolodexGo combined changeset of 2026-10-01: it is the one artifact in which ThoxEmployee team behavior and ThoxWork integrations exist together in one client codebase, implementing exactly the workflow of Section 3.2 and the handoff rule of Definition 3. It is a built, test-covered changeset that never landed in the shipped client, and it is claimed here as exactly that, no more. Second, there is one named gap: voice calling for DigitalHumans (ThoxCall) has no repository or work product in the audit scope and is deprioritized in THOX planning; it appears in this paper only in this sentence, as a roadmap item with no evidence behind it.

5 The ThoxWork Virtual Lab

The lab exists to make Sections 3's invariants falsifiable before any enterprise deployment exists to measure. It has two faces over one model. The **oracle** is a deterministic simulator

(`thoxwork_experiments.py`, Python standard library only) implementing Definitions 1 to 5 and Algorithm 1 exactly, with an independent checker that reads only the final event log and task records and never calls the admission or workflow code it verifies. The **browser lab** (the ThoxWork Virtual Lab artifact) is a view and manual driver of the same model: roster and MeshGroup panels showing pooled capacity computed by the Definition 1 formula, the Rolodex board with its four columns and a visible Denied tray, guards that reject illegal actions with the failed guard named, and experiment runners that display the oracle's summary values under the evidence notice. Where browser and oracle ever differ on the same seed, the oracle wins by definition.

Determinism. Master seed 20261104; one random generator per experiment and replicate, seeded by a fixed formula; integer ticks; sorted iteration. There are no timestamps in the evidence file, and the suite's self-test re-runs experiments and asserts byte-identical serialized results. Two full independent runs of the suite for this paper produced the same summary file, byte for byte (Section 12).

Pre-registration. The pass bars in Table 2 were fixed in the lab protocol dated 2026-11-04, before the first run. A falsified bar would have been reported as falsified; the bars were not adjusted after any result was seen.

Exp.	Invariant	Pre-registered bar
E1	I1 policy	Zero leaks in every replicate; aggregate false-block rate at most 0.01; audit coverage 1.0
E2	I2 handoffs	Chain completeness 1.0; note coverage 1.0; zero probe leaks in every replicate
E3	I3 admission	Silent overflow 0; cross-group spills 0; UNPLACEABLE recall 1.0; stranded 0; right-sized placement at least 0.99
E4	I4 review gate	Zero bypass successes in every class; zero Done-without-review; rejection integrity 1.0; legitimate completion at least 0.99
E5	I5 scale	Every grid cell completes at least 0.99 of its tasks; scaling efficiency at team size 16 at least 0.50

Table 2. Pre-registered pass bars, fixed in the lab protocol before any run.

Simulated-latency model (E5 only). Time is an abstract tick; a tick is not a millisecond of any real system. Service time per task is 10 + 2 per required resource + a uniform 0 to 4 ticks, drawn once at creation; review occupancy is 3 ticks; a coordination term of floor(0.5 × (G − 1)) ticks is added per task, where G is the MeshGroup count. Workers hold at most 2 concurrent tasks; a group's reviewer holds at most 4 concurrent reviews. Arrivals are a batch at tick 0. Reviewers in E5 are cooperative by design, because E5 measures throughput, not the gate; the gate is E4's claim. The E5 runner is event-driven; its semantics are the tick semantics above with idle time compressed, which cannot change any task's start, finish or ordering.

Workloads. E1 runs 1,000 attempts per replicate (400 legitimate; 250 direct violations; 200 cross-group; 150 post-admission escalations). E2 runs 300 tasks per replicate with reassignment depths 0 to 5, plus 200 negative probes. E3 runs 500 tasks per replicate over a demand sweep from 0.10x to 2.00x of a 120-unit pooled capacity, on a group drained to empty before each task so that queueing never confounds placeability. E4 runs 200 legitimate tasks, 300 bypass attempts in six classes, and a 200-task control arm on an otherwise identical group with the review gate switched off. E5 runs a 4 × 4 grid

(team sizes 2, 4, 8, 16; MeshGroup counts 1, 2, 4, 8) of 2,000 tasks per cell. Replicates: 20 for E1 to E4, 10 per cell for E5. Resources and capacities in the lab are abstract units (RU) over eight synthetic resource types; no mapping to real compute, storage or cost is claimed. Rosters are synthetic. The lab contains no product internals, no customer data and no real model or device specifications.

6 Evaluation

Every number in this section is read from `exp/results/summary.json`, the suite's single evidence source, and is SIMULATED. Derived ratios are CALCULATED from simulated counts. Standard deviations across replicates are 0.0 for every E1 to E4 rate, for a reason Section 7 states plainly: the mechanism under test is a deterministic rule engine with no stochastic failure channel, so the experiments are invariant checks over enumerated adversaries, not estimates of operational error rates.

6.1 E1 · Policy enforcement

Across 20 replicates, 12,000 violating attempts produced **0 leaks**: no direct, cross-group or escalation attempt was ever observed running with an unapproved resource attached. The block rate is 1.0. Of 8,000 legitimate attempts, **0 were falsely blocked** (false-block rate 0.0, against a bar of 0.01). Every attempt, accepted or rejected, appears in the event log: audit coverage 1.0. The independent checker confirms the invariant over the full log: no task ever held an unapproved resource in any running state. Bar: **met**.

6.2 E2 · Handoff integrity

Across 20 replicates, 5,000 tasks were reassigned, accumulating 15,000 handoff-chain entries to a maximum depth of 5. Chain completeness is **1.0** and note coverage is **1.0**: every entry is in sequence, correctly linked, and carries a note whose hash recomputes. The 4,000 negative probes (empty notes, whitespace-only notes, reassignment to a non-member, reassignment attempted from In Review) produced **0 leaks**: every probe was rejected and no probe mutated any chain. Bar: **met**.

6.3 E3 · Admission honesty

Across 20 replicates, all **4,000 oversized tasks were correctly denied** UNPLACEABLE (recall 1.0), with 0 silently placed, 0 cross-group spills and 0 stranded. All 6,000 right-sized tasks were placed (placement rate 1.0). Denial is issued in the same tick the task is evaluated (mean and maximum deny latency 0 ticks; see Section 7 on what that number means). Figure 3 shows the boundary exactly: at a demand of 1.00x pooled capacity, all 50 tasks per replicate are admitted; at 1.01x, all 50 are denied.

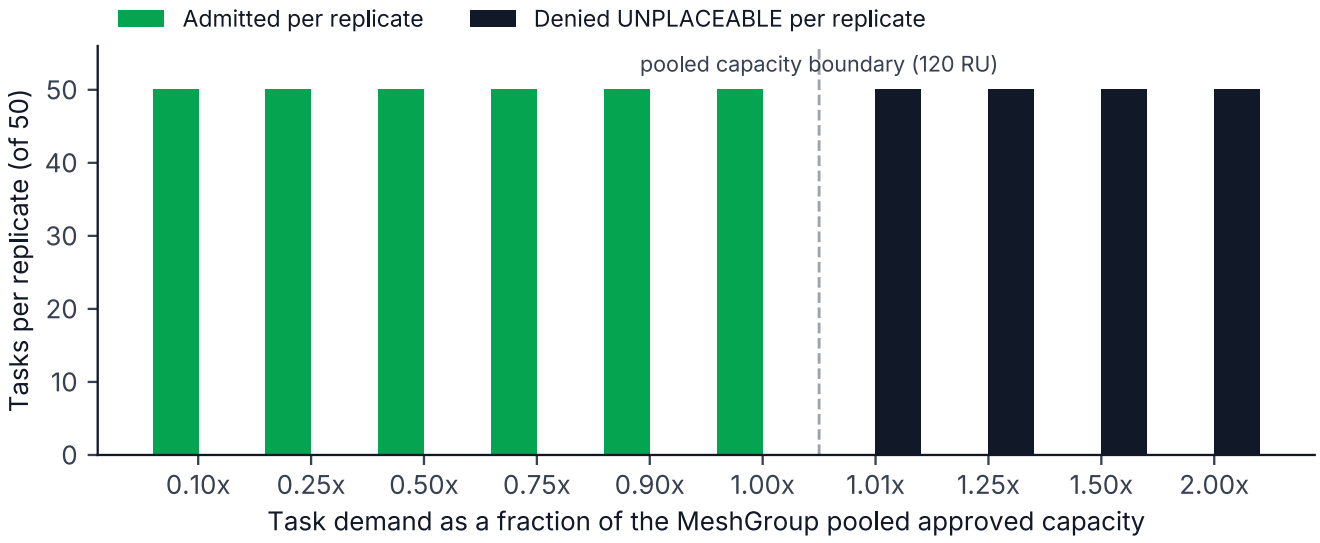


Figure 3. E3 demand sweep against a pooled approved capacity of 120 RU (per-replicate counts, identical across all 20 replicates). The step at the boundary is the Definition 5 placeability rule operating as specified. Evidence class: SIMULATED.

6.4 E4 · The review gate

Across 20 replicates, **0 of 6,000 bypass attempts succeeded**, in every class: direct finish (B1), skip review (B2), self-review (B3), wrong-role review (B4), non-member review (B5) and stale approval (B6) each record 1,000 attempts and 0 successes. Every rejected bypass left the task's state, assignee, handoff chain and review events byte-identical (rejection integrity 1.0). All 4,000 legitimate tasks completed through the full assign, check-in, submit, approve path (legitimate completion rate 1.0), and the checker finds 0 Done tasks without a legitimate review. The control arm matters as much as the result: on the identical model with the review gate switched off, 4,000 of 4,000 tasks completed without review. The gate, and nothing else in the model, is what produces the invariant, and the checker demonstrably distinguishes the two regimes. Bar: **met**.

6.5 E5 · Scale and throughput

All 16 grid cells completed **1.0** of their 2,000 tasks within budget. Throughput (tasks per 1,000 simulated ticks, cell means) ranges from 124.9 in the smallest cell (2 members, 1 group) to 9,551.2 in the largest (16 members, 8 groups). Group scaling at team size 8 is close to linear in group count (ratios 1.99, 3.70 and 6.46 at 2, 4 and 8 groups against the single-group baseline), which is the expected behavior when groups are independent pools: adding a MeshGroup adds a full team and a full resource pool, it does not subdivide one. The reviewer bottleneck is visible where the model says it should be: the share of completion time spent waiting for review is near zero at team sizes up to 4 and rises to roughly 0.20 to 0.30 at team size 16, where a single reviewer per group becomes the binding constraint. Scaling efficiency is reported and then discounted in Section 7. Bar: **met**, on the non-collapse criterion the bar actually specifies.

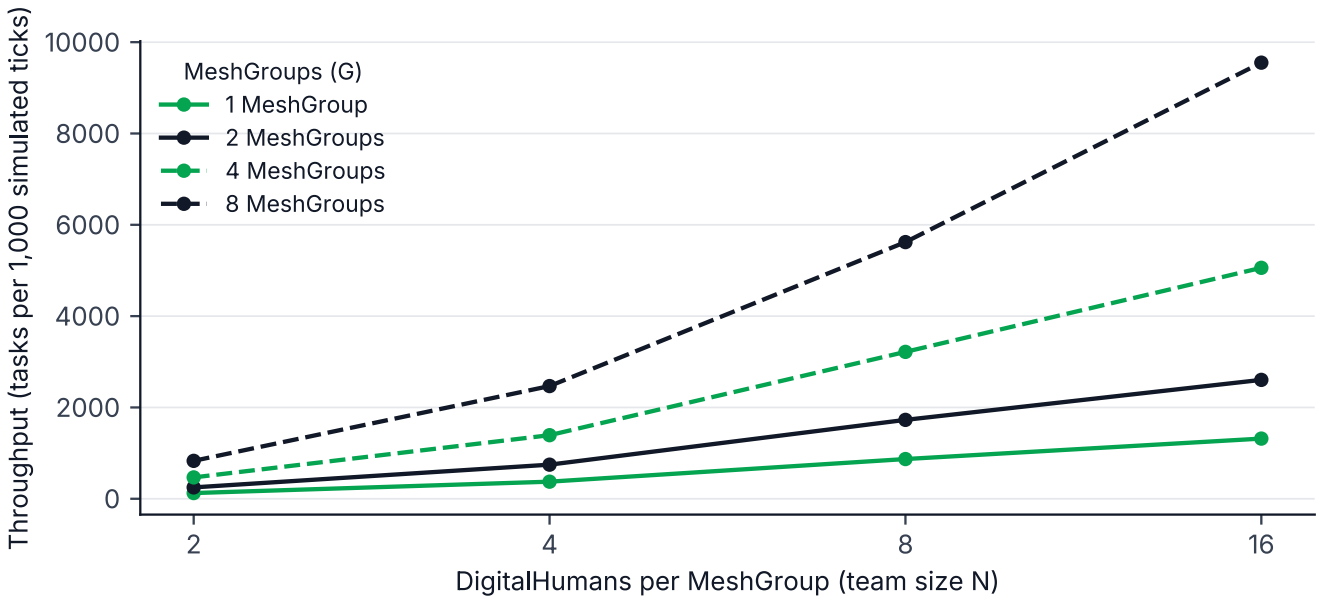


Figure 4. E5 throughput by team size and MeshGroup count (cell means over 10 replicates). Ticks are abstract simulation units. Evidence class: SIMULATED; ratios CALCULATED from simulated counts.

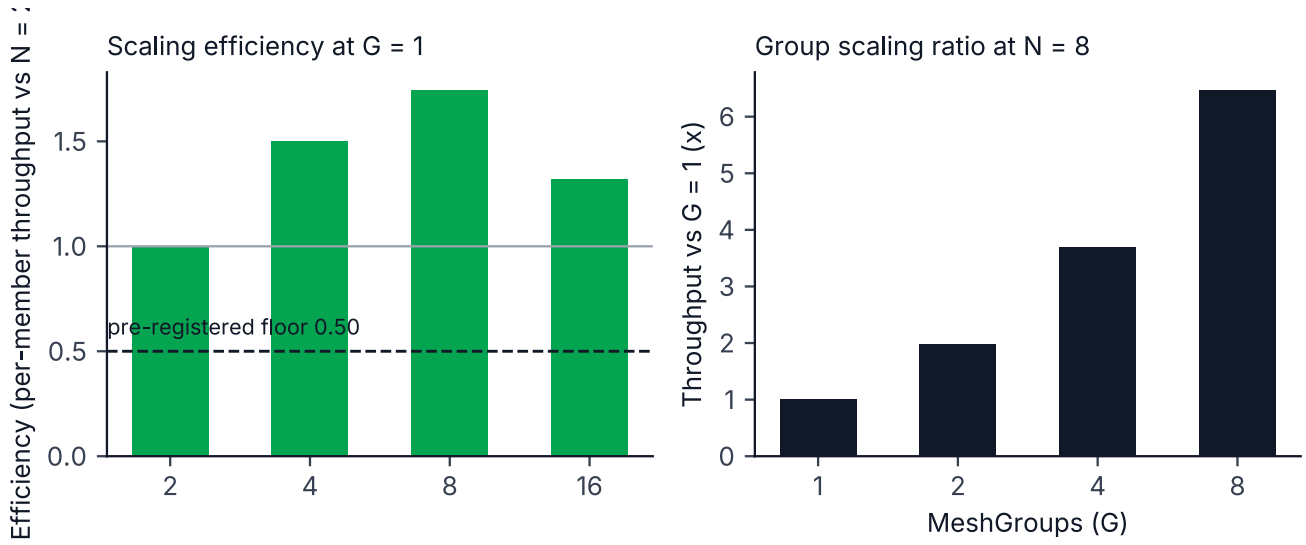


Figure 5. Left: scaling efficiency at one MeshGroup (per-member throughput relative to the 2-member baseline), with the pre-registered 0.50 floor. Right: group scaling ratio at team size 8. Values above 1.0 on the left are a model artifact analyzed in Section 7, not a speedup claim. Evidence class: SIMULATED / CALCULATED.

7 Reading the results

Four cautions are part of the results, and the paper's claims are scoped to survive them.

The E1 to E4 results are invariant checks, not error rates. A deterministic rule engine either enforces a rule against an enumerated adversary or it does not. Zero leaks over 12,000 attempts means the rules as modelled contain no path around the policy for the three attack modes tested; it does not estimate how a production implementation would behave under unenumerated attacks, implementation bugs, or

operational misuse. The zero variance across replicates says the same thing: there is no noise in the system to average over.

Scaling efficiency above 1.0 is a model artifact. The reported efficiencies (1.50 at team size 4, 1.74 at 8, 1.32 at 16, against the 2-member baseline) are superlinear, and superlinear speedup is not a claim this paper makes. Three construction choices produce it: arrivals are a single batch at tick 0, so small teams spend their early ticks queue-bound in a way that flatters larger teams' per-member figures; the only coordination cost in the model grows with group count, not with team size; and the 2-member baseline dedicates one of its two members to review rather than execution. What survives these artifacts is exactly what the pre-registered bar tested: no cell collapses, every cell completes, and per-member throughput at 16 members stays far above half the baseline. The honest reading of E5 is non-collapse under stated assumptions, plus one genuinely informative signal, the emergence of the reviewer bottleneck at team size 16, which matches the consent-fatigue and escalation-capacity concerns in the governance literature [7] and argues for reviewer capacity to be planned as a first-class resource in any real deployment.

Zero-tick denial is a definition, not a speed. E3's deny latency of 0 ticks records that admission is a synchronous decision in the model, taken in the tick of evaluation. Ticks are abstract; no wall-clock latency for any real admission implementation is claimed or implied.

The E3 cliff is the rule, drawn. The step between 1.00x and 1.01x in Figure 3 is the binary placeability comparison in Algorithm 1, not a discovered phase transition. Its value is confirmatory: the implemented rule is the specified rule, with no softening at the boundary, no partial credit, and no quiet overflow. The WAIT path of Algorithm 1 is exercised in E1 and E5 but, by the drained-group design, not in E3; `waited_then_admitted` is 0 at every E3 demand fraction, and the paper does not present E3 as evidence about queueing behavior.

8 Related work and comparison

The literature of Section 2 divides into approaches that optimize different variables. Central cloud agent platforms optimize platform scale and consolidation; single-assistant copilots optimize individual speed; open multi-agent frameworks (the LangGraph, CrewAI and AutoGen lineage, now converging with MCP and A2A [13]) optimize developer control. The comparison below is a design-position comparison, drawn from the literature survey conducted for this paper; no cited source evaluates ThoxWork, and the ThoxWork column states the position this paper argues, not a benchmark result.

Dimension	Central cloud platforms	Single-assistant copilots	Open multi-agent frameworks	ThoxWork model
Unit of work	Tenant or workflow app	Individual session	Developer-built graph or crew	Named team (MeshGroup) on shared tasks
Discovery	Central catalog in vendor cloud	None	Bring-your-own registry	Rolodex scoped per group, bounding discovery noise [1]
Where data lives	Vendor cloud by default	Vendor cloud session	Developer's deployment	Local-first: estate-primary, network optional [12]

Dimension	Central cloud platforms	Single-assistant copilots	Open multi-agent frameworks	ThoxWork model
Agent identity	Platform account	None independent of user	Integrator's problem	Named, role-holding DigitalHuman identity, distinct from owner and model [7]
Authorization	Platform roles; per-action authority rarely explicit	Inherits the user's permissions wholesale	Transport standards; policy is custom	Per-group approved resources; admission at creation and at assignment (Algorithm 1)
Human review	Configurable approval steps	User reviews own outputs	Human-in-the-loop patterns, app-specific semantics	Review as a protocol event with eligibility rules (Definition 4)
Handoffs	Platform workflow state, often conversational	Copy and paste	Conversational or graph state; loss documented [2]	Written, hashed handoff chain per reassignment (Definition 3)
Audit	Vendor-held logs	Chat history	Observability add-ons	Append-only event and evidence record in the workspace [3]
Failure containment	Platform blast radius	Single session	Hand-built circuit breakers	MeshGroup as the segmentation boundary [8]
Best-fit scale	Enterprise, central IT	Individual	Skilled development teams	Department scale first; groups compose upward by federation, not by enlarging one fleet

Table 3. Design-position comparison. The ThoxWork column is argued from the cited literature and from the model of Section 3; it is not an evaluation result.

The closest analogue remains CHAP [3], which specifies workspaces, participants, tasks, artifacts and an append-only evidence log for human-agent collaboration, with structured overrides and signed approvals. ThoxWork converges with CHAP on the workspace and evidence primitives and differs in three stated positions: its unit of authorization is the MeshGroup's approved resource set, evaluated by an admission rule with a placeability condition (CHAP does not specify resource admission); its default data posture is local-first rather than server-hosted; and its team members include DigitalHumans held out as named colleagues under an explicit colleague standard, which makes the Rolodex directory a staffing surface (who holds which role, at what tier) and not only a discovery surface. The earned-autonomy ladder of the CSA Agentic Trust Framework (Intern through Principal, with gated promotion and demotion) [8] and the identity and delegation focus areas of NIST's agent work [7] are compatible with, and more detailed than, ThoxWork's current public tiering; the paper notes both as the direction of travel for the model's authorization tiers rather than claiming conformance with either. No ratified agent-governance standard exists at writing, and this paper makes no conformance claim.

9 Product implications

PRODUCT IMPLICATIONS

For a company adopting DigitalHumans, the buying question is not which model, but which rules. The model in this paper reduces to five questions a buyer can ask any platform, including this one: Can a task touch only what its team is approved to touch, checked when it runs and not only when it is created? Is oversized work refused out loud, with a reason, instead of quietly rerouted? Does every change of hands leave a written, checkable note? Can anything reach Done without an eligible human's approval on the record? And does the evidence for all four answers live in the customer's own workspace? ThoxWork is designed so that each answer is a mechanism, not a promise, and the virtual lab in this paper exists so the mechanisms can be exercised before they are trusted.

Three implications follow for THOX's own roadmap, stated as positions rather than commitments. First, **reviewer capacity is a resource**: E5's bottleneck signal and the governance literature agree that human review is the scarcest element of the system, so MeshGroups should plan reviewer membership the way they plan any approved resource. Second, **department scale is the design point**: the orchestration evidence [1] and the comparison in Table 3 both point to bounded teams of tens, federated, over fleets of hundreds, discovered globally. Third, **the interface is the product**: Rolodex's three-verb loop (assign, check in, review) is deliberately small, because the MIT CISR and Work Trend evidence says value comes from redesigned work and team standards [4, 9], and a small, exact loop is what a team can actually standardize on. Delivery follows the positioning in THOX records: ThoxWork at the office and at home, on ThoxDevices and compatible hosts or a desktop application, with the enterprise platform itself released when THOX says it is ready and not before.

10 Security and governance considerations

This section states considerations at the level of public standards and the model's public rules. THOX's internal security configurations are out of scope.

The OWASP Top 10 for Agentic Applications (2026) enumerates the risk classes any agent platform must answer, from goal hijacking through rogue agents [11]. The ThoxWork model's public rules answer several of them structurally: per-group resource approval and double-evaluated admission bound what an agent can be made to touch (excessive agency, tool misuse); the review gate bounds what an agent can commit (improper output handling, human-review failure modes); the handoff chain and append-only event record provide the accountability substrate (repudiation). Other classes, including memory poisoning and inter-agent communication security, are not answered by anything in this paper and are named here as open obligations for any production edition. NIST's agent identity work identifies five focus areas (identification, authorization, delegation, logging and transparency, data flows) and warns that human escalation degrades under consent fatigue [7]; the model's risk-tiered review and its treatment of review as a scarce, planned resource are the design responses offered here. Persistent, named DigitalHuman personas carry a specific risk identified in the personality-pairing literature:

profile information can shape user trust and behavior, so Rolodex persona presentation must remain disclosed and factual, a constraint THOX's colleague standard also imposes [6].

11 Limitations

1. **Simulation, not deployment.** The lab proves the rules as modelled enforce the invariants. It does not prove that a production implementation enforces them. Every evaluation number is SIMULATED, and the paper's product claims are scoped by Table 1, including the platform's Coming Soon status in THOX's own copy.
2. **Enumerated adversaries only.** E1 tests three attack modes and E4 six bypass classes. An unenumerated bypass is not excluded by these results.
3. **Checker independence is definitional, not organizational.** The checker is independent code that reads only the log and final records, but it shares this paper's definitions (Definitions 3 and 4) with the engine. An error in a definition would pass both. The mitigation is the one applied here: the definitions are printed in full in Section 3 so any reader can falsify them, and the E4 control arm demonstrates the checker distinguishes gated from ungated regimes.
4. **E5 simplifications.** Batch arrivals at tick 0, cooperative reviewers, no failures, no offline members, and no members shared across groups in the grid. Real enterprise load differs on all five. The coordination cost model grows with group count only. Section 7's discounting of the efficiency figures is part of the result.
5. **Abstract units.** Capacities and demands are abstract RU. No mapping to real compute, storage, latency or cost is claimed, and the lab's resource types are synthetic.
6. **Harness simplifications in E1 to E4.** Tasks are processed sequentially (E1, E4) and concurrency ceilings are raised where concurrency is not the variable under test (E2, E3), so that capacity and concurrency never confound the policy, handoff, admission and review variables. These choices are recorded in the experiment code and do not affect the invariants' definitions.
7. **Literature transfer.** The cited studies concern other systems and populations. They are used as design warrants, and their effect sizes are not claimed for ThoxWork or its users. No peer-reviewed head-to-head comparison of local-first versus central-cloud agent-team platforms exists in the surveyed window; the local-first position is argued as a principled, data-control choice with lab support for its workflow rules, not as a proven performance winner.
8. **No customer evidence.** There are no customer deployments, testimonials or outcome metrics in this paper, because none exist in the record to cite.

12 Provenance, consent and document control

Evidence sources. Evaluation: `exp/results/summary.json` (schema `thoxwork-summary/1.0`), produced by `exp/thoxwork_experiments.py` v1.0.0, master seed 20261104, SHA-256 `f7c72f64b9b8c6f2e1663fbb014374858b2251f2aeaa7d989393a6cdb071dd79`; independently re-run and re-read by a verifier who did not produce it, byte-identical, with all five protocol bars re-checked from the

numbers (verifier report in the artifact bundle, `wp/qa/`). Product status: a repository and records audit covering 44 GitHub repositories (38 under the founder's account, 6 under the THOX organization; 43 private, 1 public) and 9 local product records, conducted read-only for this paper. Literature: a verified survey of 23 sources (recent six and twelve month windows plus labelled foundational works), from which the 15 references below are drawn. Figures 3 to 5 are generated from `summary.json` by `wp/figures.py`; Figures 1 and 2 are diagrams of the stated model.

Trade-secret boundary. This paper discloses the model's public rules and no more. It contains no internal hostnames, network details, credentials, private source code, unfiled intellectual-property particulars beyond the standing patent-pending posture, pricing, customer identities, or DigitalHuman system instructions. MeshStack internals are cited from the published record only.

Authorship and consent. Authors: Craig Ross, Chief Executive Officer, and Tommy Xaypanya, CTO and Co-Founder (corresponding author), THOX.ai LLC. Edition v1.0.1 is the public edition; edition v1.0.0 was the internal edition. This edition differs from it in edition marking and in one factual correction only: the RolodexGo Teams and Work tabs are now stated as a built, test-covered changeset that never landed in the shipped client (Section 4), where v1.0.0 presented them as part of the released client. No result, measurement, or conclusion changed. Publication proceeds under the THOX Research Paper Process v1.0 (Zenodo first, then the one-pass association sync), with each named author's consent to the exact bytes of this edition recorded in the pack's HANDOFF record. No venue submission is made by this edition.

Document control. Version v1.0.1, 2026-11-04. Changes: public edition; edition marking, plus one factual correction to the RolodexGo product-status wording in Section 4 (changeset landing status), no other content change from v1.0.0 (the internal edition). DOI: 10.5281/zenodo.23963936. Companion artifacts: interactive demo (the ThoxWork Virtual Lab), artifact bundle with experiment code, results, figures, QA report and this paper's shipped files.

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